

Project Overview

Post-Only vs Trajectory Supervision for Continual Tool-Use Learning in LLMs

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Motivation

Most of the text on the internet is final output: blog posts, answers, articles, code. But the process behind those outputs (the searching, drafting, tool usage, trial and error) is rarely captured. Humans learn from this kind of rich, step-by-step experience. LLMs, on the other hand, are mostly trained on the finished product. This creates a gap between how models are trained and how they are actually used in practice, especially when they need to keep learning over time without forgetting what they already know.

Idea

We use tool-use datasets as a natural source of process-level data. When an LLM uses a tool, the interaction produces a full trajectory: a user request, a tool call, the tool's output, and a final response. We compare what happens when a model is trained on just the final response versus the entire trajectory, in a continual learning setting where tasks arrive one after another.

Setup

We fine-tune a 7–8B open-weight model (using QLoRA for efficiency) on the API-Bank dataset, which contains dialogues across many tool domains. We split the data into sequential domain blocks to simulate a stream of new tasks. Three conditions are tested:

- **Condition A:** Train only on the final assistant response (post-only).
- **Condition B:** Train on the full trajectory (tool calls, outputs, and response).
- **Condition C:** Same as B, plus retrieval memory, experience replay, and confidence-based gating.

What We Measure

- How quickly the model adapts to new tool domains (adaptation speed).
- How much it forgets about old domains after learning new ones (catastrophic forgetting).
- How well-calibrated the model's confidence is (ECE, Brier score).
- Whether memory and replay help reduce forgetting beyond what trajectories alone provide.

Connection to the Course

The project touches on several themes from the course: continual and lifelong learning, memory systems and replay, confidence and calibration, and the general question of how richer training signals (closer to biological experience) can improve learning and retention in neural networks.